Car Price Prediction
Using Machine Learning

Project Planning & Management Documentation

This document covers the project proposal, project plan (including a Gantt chart of milestones and deliverables), task assignment and roles, risk assessment & mitigation plan, and key performance indicators (KPIs).

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1 Project Proposal

1.1 Overview

This project develops a comprehensive machine learning system to predict used-car prices from vehicle attributes such as brand, model, year, mileage, engine specifications, and transmission type. Accurate, data-driven pricing helps buyers, sellers, and dealers make informed decisions in a market where prices are otherwise negotiated with incomplete information. The project follows a full data science life cycle: data cleaning, exploratory data analysis, statistical feature validation, model development, hyperparameter tuning, and deployment-ready model persistence.

1.2 Problem Statement

Used-car pricing is currently inconsistent across regions, dealers, and individual sellers, largely because pricing decisions rely on subjective judgement or incomplete comparables. A robust predictive model that ingests a vehicle's specifications and outputs a fair market price reduces this information asymmetry and increases trust in transactions.

1.3 Objectives

- Analyze and preprocess a large-scale automotive dataset (∼432,000 records, 21 attributes) for machine learning applications.
- Identify, via statistical testing (Chi-Square, ANOVA), the features that significantly influence car pricing.
- Develop and compare six regression algorithms (Linear, Ridge, Lasso, Elastic Net, Decision Tree, Gradient Boosting).
- Select the optimal model based on R^2, RMSE, and MAE metrics.
- Produce a deployable prediction system with saved models, encoders, and preprocessing pipelines (via `joblib`).

1.4 Scope

In scope: data cleaning and outlier treatment, exploratory data analysis, feature engineering, training and tuning of six regression models, model evaluation and comparison, and persistence of all artifacts required for inference (models, label encoders, scaler, column order).
Out of scope: real-time data ingestion from live marketplaces, production-grade API/web deployment, temporal or macroeconomic pricing factors, and geographic coverage beyond the regions present in the source dataset.

1.5 Stakeholders

- **Project Team / Data Scientists** – build and validate the model.
- **Course Instructor / Evaluator** – reviews project quality and outcomes.
- **End Users (buyers, sellers, dealers)** – ultimate beneficiaries of a deployed pricing tool.

1.6 Expected Outcomes

A trained and validated Gradient Boosting / Decision Tree based regression pipeline capable of predicting car prices with high accuracy ($R^2 > 0.90$ on held-out test data), packaged with all preprocessing artifacts for straightforward integration into a Flask or FastAPI service.

2 Project Plan

2.1 Project Timeline (Gantt Chart)

The project was executed over an 8-week period, organized into six phases that mirror the technical workflow described in the methodology (data preprocessing, EDA, feature engineering, model development, evaluation, and deployment packaging).

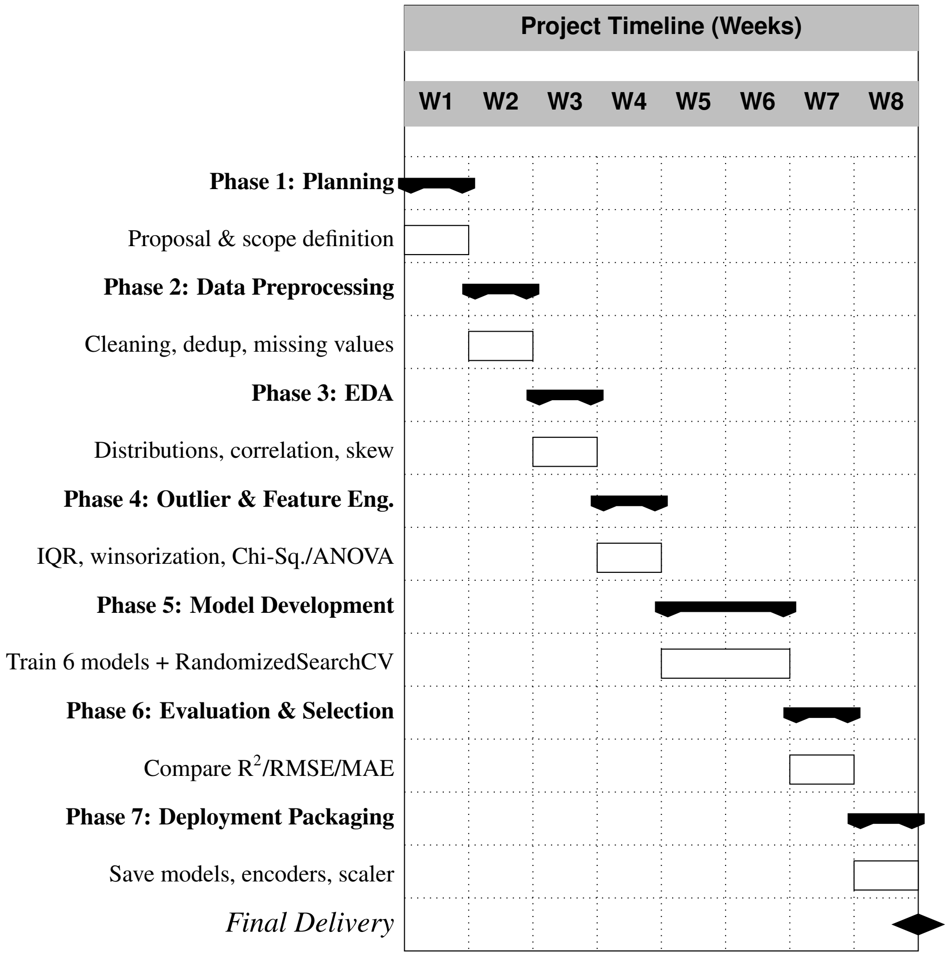

Figure 1: Project Gantt chart showing phase-level scheduling across 8 weeks.

2.2 Milestones

Milestone	Description	Target Week	Status
M1	Project proposal approved, dataset acquired	Week 1	Completed
M2	Clean dataset (nulls, duplicates, irrelevant columns removed)	Week 2	Completed
M3	EDA report complete (distributions, correlation matrix, skewness)	Week 3	Completed
M4	Outlier treatment finalized & statistically significant features confirmed	Week 4	Completed
M5	All six regression models trained and tuned via RandomizedSearchCV	Week 6	Completed
M6	Best model selected (Gradient Boosting / Decision Tree)	Week 7	Completed

Milestone	Description	Target Week	Status
M7	Models, encoders, scaler, and column order persisted with `joblib`	Week 8	Completed

2.3 Deliverables

- Cleaned and preprocessed dataset.
- EDA report (histograms, count plots, correlation heatmap, skewness table).
- Statistical significance report (Chi-Square and ANOVA test results).
- Six trained regression models with tuned hyperparameters.
- Model comparison report (R^2, RMSE, MAE) with bar-chart visualization.
- Serialized artifacts: `best_model.pkl`, individual model `.pkl` files, `label_encoders.pkl`, `minmax_scaler.pkl`, `columns.pkl`.
- Final project documentation (this package).

2.4 Resource Allocation

Resource	Purpose	Allocation
Compute (Google Colab)	Model training, hyperparameter search	Shared, on-demand during Phases 5–6
Dataset storage (Google Drive)	Central dataset and artifact storage	Continuous, all phases
Data Scientist(s)	Cleaning, EDA, modeling, evaluation	Full-time, Weeks 1–8
Python / scikit-learn / XGBoost stack	Core modeling libraries	All phases
Documentation time	Reports, charts, final write-up	Weeks 1, 7–8

3 Task Assignment & Roles

3.1 Team Roles and Responsibilities

Role	Responsibilities	Key Deliverables
Project Lead / Data Scientist	Overall coordination, methodology design, final model selection	Model comparison report, final recommendation
Data Engineer	Data ingestion (Google Drive mount), cleaning, missing-value and duplicate handling	Cleaned dataset
EDA Analyst	Distribution analysis, correlation matrix, skewness assessment, visualizations	EDA notebook and charts
Feature Engineer	Outlier detection/treatment (IQR, winsorization), Chi-Square and ANOVA testing, encoding, scaling	Feature validation report, encoded dataset

Role	Responsibilities	Key Deliverables
ML Engineer	Implementation and tuning of the six regression models via RandomizedSearchCV	Trained models with best hyperparameters
Evaluation & Deployment Lead	Metric computation (R^2, RMSE, MAE), model persistence with `joblib`, deployment readiness	Saved model artifacts, evaluation report

3.2 RACI Matrix

Task	R	A	C	I
Data cleaning & preprocessing	Data Engineer	Project Lead	EDA Analyst	Team
Exploratory Data Analysis	EDA Analyst	Project Lead	Feature Engineer	Team
Outlier treatment & statistical testing	Feature Engineer	Project Lead	ML Engineer	Team
Model training & hyperparameter tuning	ML Engineer	Project Lead	Feature Engineer	Team
Model evaluation & selection	Evaluation Lead	Project Lead	ML Engineer	Team
Model persistence & packaging	Evaluation Lead	Project Lead	Data Engineer	Team
Final documentation	Project Lead	Project Lead	All	Instructor

R = Responsible, A = Accountable, C = Consulted, I = Informed.

4 Risk Assessment & Mitigation Plan

4.1 Risk Register

Risk	Likelihood	Impact	Mitigation Strategy
Missing data in key columns (e.g., `complectation`, `super_gen_name`)	Medium	Medium	Drop columns with excessive missing values; drop remaining null rows after confirming volume impact is acceptable.
Outliers skewing model training (e.g., `displacement`, `power`)	High	High	Two-stage treatment: IQR-based detection followed by winsorization at 1st/99th percentiles, validated with boxplots.
Overfitting in tree-based models (Decision Tree, Gradient Boosting)	Medium	High	Hyperparameter tuning via RandomizedSearchCV with 3-fold cross-validation; monitor train/test R^2 gap.

Risk	Likelihood	Impact	Mitigation Strategy
High-cardinality categorical features (e.g., 2020 unique `model` values)	Medium	Medium	Label encoding instead of one-hot encoding to control dimensionality; statistical testing to prune weak features.
Multicollinearity among numerical features (e.g., `super_gen_year_from/to` vs. `year`)	Medium	Medium	Correlation heatmap review; regularized models (Ridge, Lasso, Elastic Net) included as a check on linear-model stability.
Irreproducible results across runs	Low	Medium	Fixed `random_state=42` across all splits and models; all preprocessing artifacts persisted with `joblib`.
Data drift / lack of temporal and economic factors	Medium	Medium	Documented as a known limitation; recommend periodic retraining and monitoring in future work.
Limited geographic representation in dataset	Low	Medium	Flagged as a scope limitation; region treated as a categorical feature rather than assumed uniform.
Compute/environment failure (e.g., Colab session loss)	Low	Medium	Google Drive mounted for persistent storage; all trained models and encoders saved incrementally.

4.2 Risk Monitoring

Risks are reviewed at the end of each project phase (see Gantt chart, Section 2.1). Any newly identified risk is logged in the risk register above with an assigned likelihood, impact, and mitigation owner before the next phase begins.

5 Key Performance Indicators (KPIs)

5.1 Model Performance KPIs

KPI	Definition	Target / Result
R^2 Score	Proportion of variance in price explained by the model	Target ≥ 0.90; achieved 0.9376 (Decision Tree)
RMSE	Root Mean Squared Error, penalizes large prediction errors	Minimize; best result 291,939 (Decision Tree)
MAE	Mean Absolute Error, robust average error magnitude	Minimize; best result 173,310 (Decision Tree)
Model improvement over baseline	% R^2 gain of best model vs. Linear Regression baseline	Achieved $\sim$28.9% relative gain ($0.7273 \rightarrow 0.9376$)
Cross-validation stability	Variance of R^2 across CV folds during RandomizedSearchCV	Low variance across 3-fold CV

5.2 Process & Delivery KPIs

KPI	Definition	Target / Result
Schedule adherence	% of milestones completed within planned week	Target 100%; achieved 100% (7/7 milestones)
Data quality improvement	Reduction in missing values and duplicates after cleaning	Duplicates reduced to 0; nulls eliminated
Outlier reduction rate	% reduction in flagged outliers after winsorization/IQR removal	Reduced dataset from 432,227 to 333,801 valid rows
Feature validation coverage	% of categorical features statistically tested (Chi-Square/ANOVA)	100% (11/11 categorical features tested)
Deployment readiness	Completeness of saved artifacts required for inference	100% (models, encoders, scaler, column order all saved)

5.3 Future Monitoring KPIs (Post-Deployment)

- **System uptime** – % availability of the prediction service once deployed via Flask/FastAPI.
- **Response time** – average latency (ms) for a single price prediction request.
- **User adoption rate** – number of active users/dealers querying the model per period.
- **Prediction drift** – periodic comparison of live prediction error against the original test-set benchmark, triggering retraining if it exceeds an agreed threshold.